

Tier 1 roadmap (SCOUT / VECTOR) — living checklist

Location: `sports/documents/tier_1_roadmap.md` (alongside other sports-facing notes; not a duplicate of theory memos in `5-Manuscript`.)

Notebook: `sports/535_sports_tier_1.ipynb`

Purpose: Single place for the mechanical pipeline, the `df` column contract after each stage, and **CELL 0** switches. Update this file as CELL 4+ and modeling land.

Contract (what “must be true” for implementation): this file + the **module docstrings / function contracts** in `sports_pipeline/tier1_mechanism_vars.py` and `sports_pipeline/panel_build.py` (especially LOO / `poolq_loo` behavior). Manuscripts in `5-Manuscript` are **design intent and exposition**, not the execution contract unless we explicitly mirror a rule here.

Related theory (`5-Manuscript` , stay at source — link, don’t copy):

- `Vector_to_Scout_Tier1_Modeling_Direction.md` — main Tier 1 modeling direction.
- `tier_1_model.md` — model layer notes.
- `Tier1_Briefing_Outline.md` — outline answering Alex’s May 5 briefing guidance: model components, assumptions, fitting plan, and turning point.
- `Tier1_Narrative_Outline.md` — current narrative flow: competing local forces, L -first Tier 1 model, Λ , fitting plan, variable domains, and reference addenda.
- `2026_0430_Paper7_feedback.md` — theory vs minimal model; scarcity.

Optional depth (broader VECTOR notes): `Vector_Master_Theory_and_Modeling_Notes*.md` , `Vector_Questions_and_Modeling_Thoughts.md` , `Vector_Evans_Reaction_and_Theoretical_Expansion_Notes.md` — pull excerpts into this roadmap only when a line becomes a **coded rule** or sample definition.

Where We Have Been / Where We Are Going

Keep this section open while working in `535_sports_tier_1.ipynb` . Green items are done/validated; black/default items are next work or open decisions.

Done / Validated

- **DONE — Mechanical notebook structure:** `535_sports_tier_1.ipynb` now follows the named-cell pattern: markdown header, then code cell with matching first comment line.
- **DONE — CELL 0 switchboard:** run gates and main knobs live in CELL 0 (`RUN_CELL1` — `RUN_CELL4` , `PERF_METRIC` , `PRIMARY_POOL_MODE` , `COMPUTE_WEIGHTED_CROWDING` , etc.).
- **DONE — 530 source-of-truth workflow:** exact 530 reproduction requires running `530_sports_pipeline.ipynb` from scratch with `use_prebuilt_panel_csv=False` and `export_panel_after_run=True` , then loading the refreshed `datasets/mbb/player_season_panel_530.csv` in 535.
- **DONE — Reproduced the reduced-form phenomenon:** with the refreshed 530 panel, `poolq_loo` binned bars reproduce the inverted-U / top-bin-drop pattern.

- **DONE — Plot-style switch:** CELL 4 supports `CELL4_EDA_PLOT_STYLE = "poolq_line"` vs `"bins_bars_520"` so we can compare quality-scale lines with 530-style bin-index bars.
- **DONE — Tier 1 mechanism variables:** CELL 3 creates `congestion_quality`, `congestion_crowding`, and, when enabled, `congestion_crowding_weighted`.
- **DONE — Weighted crowding behavior made explicit:** when `PRIMARY_POOL_MODE = "crowding"` and `COMPUTE_WEIGHTED_CROWDING = True`, CELL 4 plots `congestion_crowding_weighted`; when the boolean is false, it plots plain `congestion_crowding`.
- **DONE — First Tier 1 EDA pass:** binned bars for `congestion_quality`, plain `congestion_crowding`, and weighted crowding have been compared against the original `poolq_loo` pattern.

Next / TODO

- TODO — Decide the first empirical proxy for L_{ijt} : likely start with `poolq_loo` / `congestion_quality`; treat `congestion_crowding`, `congestion_crowding_weighted`, and `minutes` as later decomposition diagnostics unless they earn a clearer first-pass role.
- TODO — Fix or demote the right-panel dashed model curve: current clipped LPM / PD-style curves can be misleading; print unclipped predictions and coefficients before trusting the curve.
- TODO — Add a cleaner CELL 5 regression block: start with $Y \sim L + L^2 + A$ where L is first proxied by `poolq_loo` / `congestion_quality`; then add one decomposition diagnostic at a time (`congestion_crowding`, weighted crowding, `minutes`, interactions) after the minimal shape is clear.
- TODO — Decide binning defaults by purpose: use `equal_width` when matching a 530 export slug; use `quantile` first for Tier 1 mechanism diagnostics so bins are populated.
- TODO — Add export behavior for 535 figures/tables only after the Tier 1 specs settle enough to save reproducible outputs.
- TODO — Update theory language after EDA: distinguish reduced-form `poolq_loo` from mechanism variables (`quality`, `crowding`, weighted crowding).

Pipeline (numbered)

1. **Load the panel** — `load_panel(CFG)` → `df` from `player_season_panel_530.csv` at **workspace root** `datasets/mbb/` (via `sports_pipeline.paths.panel_530_csv()`). For exact 530 reproduction, refresh this CSV first by running **530** from scratch with `use_prebuilt_panel_csv=False` and `export_panel_after_run=True`.
2. **Set `perf` from a metric** — e.g. PPM via `assign_perf_from_metric(df, PERF_METRIC)` with `PERF_METRIC` defined in **CELL 0**.
3. **Recompute legacy `poolq_loo` (and `poolq_sq`)** — `recompute_teammate_loo_pool_quality(...)`.

Why recompute instead of trusting the CSV?

- LOO pool quality must be built off the `perf` you just chose; a saved `poolq_loo` may reflect an older or different `perf` definition.
- Optional **winsor** on `poolq_loo` via `CFG.poolq_winsor_quantiles` is applied in this step when set.

Comparison: `poolq_loo` (legacy) and `congestion_quality` (Tier 1) can differ slightly when some roster rows have missing `perf`; the Tier 1 helper uses only rows with valid `perf` in the `(team_id, season)` group for denominators. See `sports_pipeline/tier1_mechanism_vars.py` module docstring.

4. **Add Tier 1 mechanism columns** — `add_tier1_mechanism_variables(...)`:
 - `congestion_quality` — LOO **mean** teammate `perf` (valid Q for Tier 1).
 - `congestion_crowding` — LOO **sum** teammate `perf`.

- `congestion_crowding_weighted` — optional weighted variant (minutes × `perf` logic); gated by **CELL 0** `COMPUTE_WEIGHTED_CROWDING` .
- If `MIN_MINUTES_TIER1` is set in **CELL 0**, rows below that **minutes** threshold are dropped **before** group stats.

5. **CELL 4 — Tier 1 EDA** (notebook): On a sample with valid `PRIMARY_POOL_MODE` pool column, `congestion_crowding`, `minutes`, `Y_draft`: descriptive `describe()`; **binned** mean draft vs Q using `assign_poolq_bin_labels` with `CFG.ventiles` and `CFG.poolq_binning`; numpy **LPM** $Y \sim 1 + Q + Q^2$; multivariate linear $Y \sim 1 + Q + Q^2 + C + \text{minutes}$ when Q is quality (if Q is crowding, **omit duplicate C**); **PD-style** predicted curve over a Q grid with C and minutes at sample medians (linear approx, clipped to [0,1]). When `PRIMARY_POOL_MODE="crowding"` and `COMPUTE_WEIGHTED_CROWDING=True`, the binned Q variable is `congestion_crowding_weighted`; otherwise it is plain `congestion_crowding` .

`df` column contract (Tier 1–relevant)

After steps 1–3 (after notebook CELL 2):

- Required for assertions in the notebook: `Y_draft`, `poolq_loo`, `minutes` .
- Always present after recompute (when inputs allow): `perf`, `poolq_loo`, `poolq_sq` .
- Plus **all other columns** from the 530 CSV (identifiers such as `athlete_id`, `team_id`, `season`, and any merges from 530).

After step 4 (after notebook CELL 3):

- Same as above, **plus**: `congestion_quality`, `congestion_crowding`, and usually `congestion_crowding_weighted` (or NaNs if weighted path is off / invalid inputs).
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CELL 0 — variables / boolean switches (current)

Kind	Name	Role
Run gate	<code>RUN_CELL1</code>	Environment · imports · panel path
Run gate	<code>RUN_CELL2</code>	Load panel · PPM · legacy <code>poolq_loo</code>
Run gate	<code>RUN_CELL3</code>	Tier 1 mechanism variables
Run gate	<code>RUN_CELL4</code>	Tier 1 EDA (bins, LPM, PDP-style Q grid)
Knob	<code>PERF_METRIC</code>	Passed to <code>assign_perf_from_metric</code> (default <code>"ppm"</code>)
Knob	<code>CELL4_MATCH_530_VENTILE</code>	<code>True</code> = 530-style <code>poolq_loo</code> ; <code>False</code> = Tier 1 mechanism variable path
Knob	<code>CELL4_EDA_PLOT_STYLE</code>	<code>"poolq_line"</code> vs <code>"bins_bars_520"</code>
Knob	<code>MIN_MINUTES_TIER1</code>	Optional row filter before Tier 1 group stats; <code>None</code> = full sample
Knob	<code>PRIMARY_POOL_MODE</code>	<code>"quality"</code> vs <code>"crowding"</code> for Tier 1 CELL 4 path
Knob	<code>COMPUTE_WEIGHTED_CROWDING</code>	Whether to build <code>congestion_crowding_weighted</code> ; when <code>PRIMARY_POOL_MODE="crowding"</code> , <code>True</code> makes CELL 4 plot weighted crowding

Pattern matches `540_tenure_pipeline.ipynb` CELL 0: one switchboard, re-run after changes.

Next (placeholder)

- CELL 5+: proper logit / `statsmodels`; cluster SEs; season or team FE; richer PDP; export figures to `CFG.exports_dir` if you want parity with 530 PNG exports.

Changelog

Date	Note
2026-05-07	Added <code>Tier1_Narrative_Outline.md</code> to related theory and updated TODOs to reflect the current L -first Tier 1 framing with crowding/minutes as decomposition diagnostics.
2026-05-06	Added manuscript briefing outline link for Alex's May 5 guidance: model components, assumptions, fitting plan, and Q^* turning point.
2026-05-05	Added "Where We Have Been / Where We Are Going" dashboard; documented 530 source-of-truth workflow and weighted-crowding CELL 4 behavior.
2026-05-06	CELL 4 in 535: binned draft vs Q, quadratic LPM, C + minutes controls, PDP-style Q grid; <code>RUN_CELL4</code> in CELL 0.
2026-05-06	<code>paths.project_root</code> = Ivy_Net workspace root so <code>datasets/mbb/</code> matches 530 output (not <code>sports/datasets/mbb/</code>).
2026-05-05	File lives under <code>sports/documents/</code> ; contract vs theory section + links to <code>5-Manuscript</code> .
2026-03-31	Initial file: pipeline steps, recompute rationale, <code>df</code> contract, CELL 0 table.